DSA Experiment 9:Represent graphs and generate minimum spanning trees using Prim's algorithm.

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```
#include <stdio.h>

#define V 5

int main()
{
    int graph[V][V] = {
        {0, 2, 0, 6, 0},
        {2, 0, 3, 8, 5},
        {0, 3, 0, 0, 7},
        {6, 8, 0, 0, 9},
        {0, 5, 7, 9, 0}
    };

    int visited[V] = {0};
    int edges = 0;
    int min, u, v, total = 0;

    visited[0] = 1;  // Start from vertex 0

    printf("Edge  Weight\n");

    while (edges < V - 1)
    {
        min = 999;

        for (int i = 0; i < V; i++)
        {
            if (visited[i] == 1)
            {
                for (int j = 0; j < V; j++)
                {
                    if (visited[j] == 0 && graph[i][j] != 0)
                    {
                        if (graph[i][j] < min)
                        {
                            min = graph[i][j];
                            u = i;
```

```
                    v = j;
                }
            }
        }
    }

    printf("%d - %d   %d\n", u, v, min);

    visited[v] = 1;
    total = total + min;
    edges++;
  }

  printf("Total MST Weight = %d\n", total);

  return 0;
}
```

Output

```
Edge  Weight
0 - 1   2
1 - 2   3
1 - 4   5
0 - 3   6
Total MST Weight = 16
```